

I am reading this in 2025 so I have hindsight.

1) The paper introduces mlps in the network so instead of sliding a linear kernel they slide an mlp over the features.

2) The Paper also introduces GAP (global avg pooling)

On the topic of 1. They say that convolution filters are just "generalized linear models" so they propose to replace the GLM with a more potent non-linear function approximator... MLP.

GLM works well when the latent space is linearly separable, but if the data that captures some concepts can be in a highly non-linear manifold.

On the topic of 2. They add this GAP for a few reasons, one is to add regularization to the network, cuz when you did linear layers you needed to use lots of Dropout. & also these gaps represent a features "confidance"

So one we are in our network many of our concepts live on a non-linear manifold (all car images but from different angles live on some manifold)

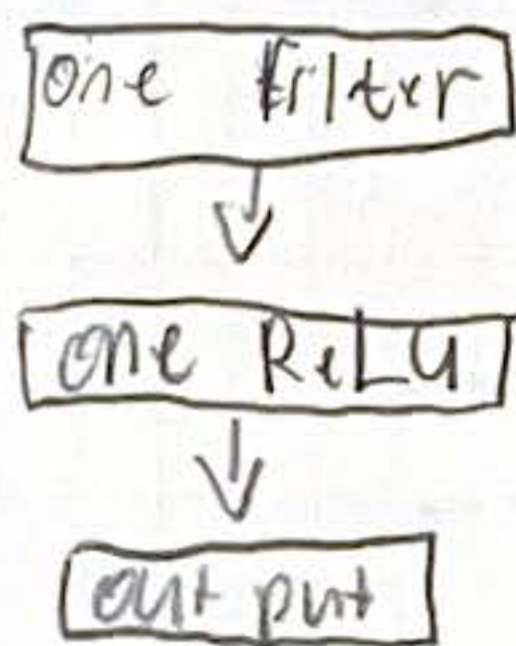
So to separate things in this space we would need very many convolutions followed by non-linearities since convolutions are GLM's

But if we introduce these mlps they can represent these splits more efficiently.

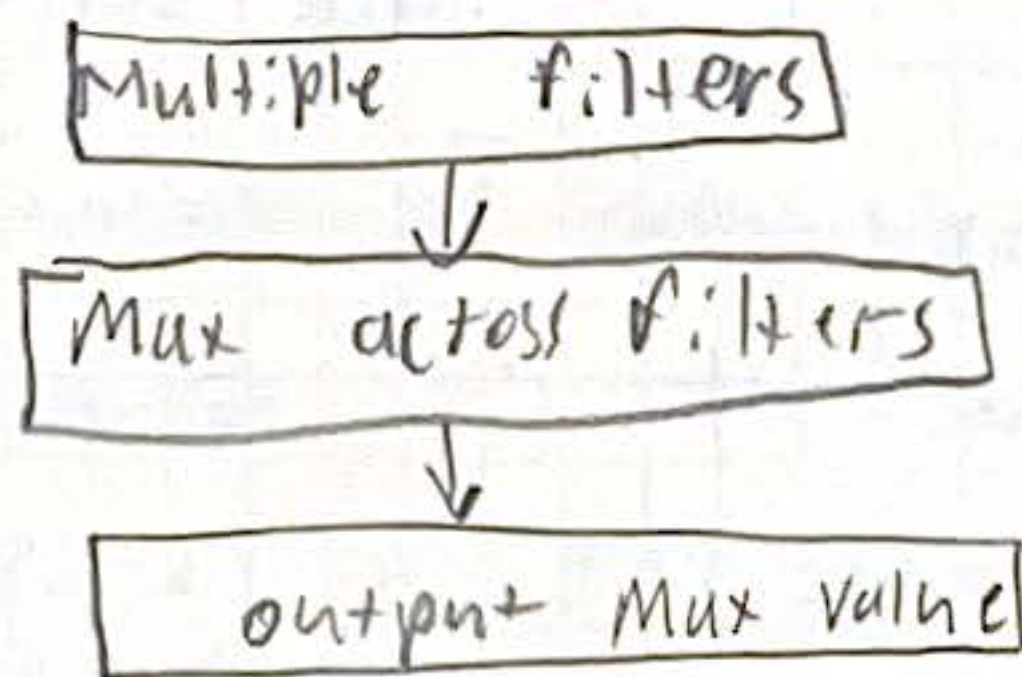
So they note of a paper before this one that did "Maxout" network

where:

Regular Network



Maxout Network



Maxout can be a piecewise approximator

Since maxout can learn convex functions it's an intermediate between that & what they propose here since MLP can approximate all functions.

Now they will explain in detail each of these parts of the paper:

- MLP convolution
- GAP
- Network in Network structure

MLP Convolution:

So they say if we have no priors about the distribution of the latent space then we should use a universal function approximator so it can capture all of the info.

They explain what it looks like by...

an MLP convolution is the same as a cascade of 1×1 convolution NOT to be confused with a single 1×1 conv.

Input $\rightarrow 1 \times 1$ conv $\rightarrow$ output Linear

Input $\rightarrow 1 \times 1$ conv + ReLU $\rightarrow 1 \times 1$ conv + ReLU
 $\rightarrow 1 \times 1$ conv + ReLU $\rightarrow$ output

Single 1x1 conv GLM

$$\text{output} = W \times \text{input} + b$$

MLP conv

$$h1 = \text{ReLU}(W1 \times \text{input} + b1)$$

$$h2 = \text{ReLU}(W2 \times h1 + b2)$$

$$\text{output} = \text{ReLU}(W3 \times h2 + b3)$$

GLOBAL AVERAGE POOLING:

They note the previous thing of vectoring the extracted feature maps then passing to linear layers for classification was prone to overfitting.

So they propose to avg across a feature map. Since it's just avg pool across the entire map there is one parameter to optimize so it acts as a regularizer. Additionally, it forces the feature maps to act as a sort of confidence about the specific feature.

Experiments

- they perform at the best or near the best on many image classification datasets.
(2014) state of the art.

To test the regularization benefits of GAP they test:

- mlpconv + Fully connected
- mlpconv + FC + Dropout
- mlpconv + gap

and mlpconv + gap was the best followed by mlpconv + FC + Dropout.